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H. Dip. Computer Science 2024

Final Project Report

Proof of Concept: **Rehabilitation Dashboard with Integrated Activity Tracker Data and Bluetooth Movement Sensor Metrics, with exploration of Retrieval Augmented Generated Classification of Movement**

Student: *Conor Leonard*

Student Number: *20109017*

Declaration

I declare that the work which follows is my own, and that any quotations from any sources (e.g. books, journals, the internet) are clearly identified as such by the use of 'single quotation marks', for shorter excerpt and identified italics for longer quotations. All quotations and paraphrases are accompanied by (date, author) in the text and a fuller citation is the bibliography.

ChatGPT (paid subscription) was used extensively in this project as a reference tool, for idea generation, for problem solving and data analysis, for code suggestions, for error interpretation, for preliminary draft of sample paper text (all then personally reviewed and edited) and for assistance on tasks such as deployment and preparation of references. Gemini was used for image generation of the architecture diagram. Links to interactions are provided at the end of this document.

I have not submitted the work represented in this report in any other course of study leading to an academic award.

Student..... Date

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As ever, I am indebted to my family for their tolerance of my endless educational endeavours. To quote what my children once told their friends many years ago: "My dad must be really thick; he had to go to university three times".

Never say never again.

Conor Leonard

Diploma of The College of Occupational Therapists, University of Ulster, 1985
Master of Science, Community Health, TCD, 1991
Master of Business Administration, Healthcare Management, RCSI/UCD, 2006
H Dip Data Analytics, National College of Ireland, 2018

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Abstract

This project presents a proof-of-concept (PoC) for web-based rehabilitation dashboard integrating wearable activity-tracker device data with Bluetooth Low Energy (BLE) movement sensing for timely movement detection and objective activity measurements, presented in a React front-end for clinician ease-of-use. It is based on the author's experience in clinical rehabilitation of older persons, which requires mobilisations whilst minimising the risk of falls. The project combines 'Fitbit' activity and physiological metrics pulled from the Fitbit Cloud via an Application Programme Interface (API) and combines these with real-time Bluetooth-transmitted sensory gyroscopic movement telemetry readings captured via a Raspberry Pi acting as an edge gateway. A Node.js (Hapi) backend aggregates and combines time-series data from both sources, with persistence achieved via a Google Firestore database. Cloudinary is used for images. The React frontend renders the combined data, presenting simulated ward level and individual patient data using ReChart.

The system potentially provides early warning of patient movements that would not be identified by a standard wearable activity tracker. Early exploration of RAG analysis suggests movement trends for an individual can be customised, based on observed movement data sets. Implementation involved a multi-layer architecture, integrating tests for device connectivity tests and data flows.

The project provides basic proof of the feasibility of a distributed, per-patient deployment model, wherein each patient could be supported by a

- wearable activity tracker device,
- mobile phone and subscriber account for synchronization
- dedicated edge device for BLE capture.
- multi-functional BLE movement sensor

The results demonstrate technical feasibility whilst recognising real-world constraints. Clinical viability would of course require considerable further study.

Introduction

The author has nearly 40 years experience in the rehabilitation of older people in both clinical and managerial roles. This project was based in a desire to enhance access to objective rehabilitation data and provide potentially superior early warning of patient movement as a possible means of preventing falls.

Ireland is undergoing a significant demographic transition characterised by rapid population ageing. Projections from the Central Statistics Office (CSO) indicate that the number of people aged 65 years and over will exceed one million by 2030 (CSO, 2024a). Of particular significance is the expansion of the “oldest old” cohort: the population aged 85 years and over is projected to increase dramatically, from just over 100,000 in the late 2020s to approximately 389,000 by 2057, almost a fourfold rise (CSO, 2025).

This disproportionate growth in the 85+ group is critical, as it is closely associated with higher levels of frailty, multimorbidity, and demand for complex health and social care services. Older persons represent a disproportionately high user group of acute hospital services in Ireland, a pattern explicitly recognised within the Health Service Executive National Clinical Programme for Older People, which identifies them as “regular users of acute hospital services” requiring integrated pathways spanning emergency, inpatient and rehabilitation care (HSE, 2012).

This high utilisation is closely linked to frailty, multimorbidity and functional decline, with evidence indicating that between 20% and 50% of older adults presenting to hospital exhibit frailty, rising to as high as 80% in specialist geriatric settings (HSE, 2012). Importantly, hospitalisation itself is associated with significant functional deterioration, with approximately 35% of adults aged 70 and up to 65% of those aged 90 experiencing functional loss by discharge (HSE, 2012).

This creates a substantial downstream requirement for rehabilitation. Evidence indicates that around 60% of older hospital patients present with pre-existing functional decline, and a large proportion fail to regain baseline function without structured intervention (ISPGM, 2024). Taken together, these findings support a reasonable service-planning estimate that between 30% and 60% of older hospitalised patients will require some level of rehabilitation,

ranging from early supported discharge and community-based therapy to inpatient geriatric rehabilitation.

Every rehabilitation setting aims to provide effective services. A key challenge for clinicians (and hospital management) is to be confident that patients are achieving sufficient daily movement whilst actively reducing the risk of falls (WHO, 2008). The most serious falls result in hip fractures, which are deemed 'serious reportable incidents' which hospitals must report to authorities. These injuries are associated with major clinical and economic consequences including long hospital stays, loss of independence and very high mortality: one in three patients will not survive a full year. In Ireland, hip fractures are estimated to cost €296M annually, at just under €12K per episode, including an average of 17 days of acute hospital admission (Ferris et al., 2022).

Rehabilitation of older adults requires consistent mobilisation to prevent deconditioning, preserve functional independence, and reduce falls risk. In practice, however, monitoring of patient activity is intermittent and largely dependent on staff observation and documentation. This introduces variability and subjectivity particularly outside typical weekday therapy hours.

One possible solution to reduce subjectivity is the use of wearable activity trackers and movement sensors. This project attempts to address this issue by developing a proof-of-concept system that continuously captures and presents patient activity using a combination of wearable devices and movement sensors suited to an indoor environment.

The central hypothesis is that integrating multi-source data into a clinician-facing dashboard can improve patient situational awareness, help with falls prevention and support objective rehabilitation measurement.

RAMP

The project followed a RAMP (Research, Analysis, Modelling, Planning) approach. Research informed technology selection and constraints; analysis defined system requirements; modelling established the architecture and interfaces; and planning used iterative sprints to progressively integrate components. Development began with simulated data to stabilise interfaces before introducing real data sources. The architecture leveraged the approaches as introduced in the earlier modules of the SETU HDip course and associated projects e.g. Node, Hapi, Cloudinary, IoT.

Research

Wearable sensors and Bluetooth Low Energy (BLE) technologies have increasingly been studied for indoor positioning and movement monitoring in healthcare environments (Faragher and Harle, 2015; Liu et al., 2016). Studies have shown that integrating wearable sensor data with indoor localisation can produce meaningful activity and location features that help identify patterns of recovery and risk in rehabilitation populations, although these insights are not typically presented through clinician-focused dashboards (Ramezani et al., 2019). Bluetooth Low Energy (BLE) beacon systems provide a low-cost and low-power approach to estimating proximity and movement indoors where Global Positioning Satellite (GPS), used by wearables and/or their associated mobile phone, is not suitable (Faragher and Harle, 2015). Studies suggest BLE signals can support detection of transitions such as bed exit, room movement and walking patterns, making them suitable for clinical monitoring environments (Faragher and Harle, 2015; Patel et al., 2012).

Wearable Activity Monitoring in Rehabilitation

Wearable activity trackers have been widely studied as tools for monitoring physical activity, heart rate, sleep, and engagement in rehabilitation programmes (Patel et al., 2012; Wright et al., 2017). These devices provide continuous, longitudinal data that can supplement episodic clinical assessments. Evidence suggests that step counts and activity duration correlate with recovery trajectories and can be used to identify changes in functional status over time (Liu et al., 2016).

Limitations of GPS for Indoor Monitoring

Global Positioning System (GPS) technologies are unreliable in indoor environments due to signal attenuation, non-line-of-sight conditions, and multipath interference (Blunck et al., 2009). Studies consistently report poor accuracy indoors, making GPS unsuitable for room-level or bed-level localisation (Faragher and Harle, 2015; Kim Geok et al., 2020; Kunhoth et al., 2020). Consequently, alternative approaches may be required for settings such as hospitals (Hailu et al, 2024, Jiang, H. 2023)

Falls prevention

Existing falls-prevention solutions include pressure pad alarms, camera-based monitoring systems, smart beds and wearable motion-sensing devices, reflecting a broader shift toward sensor-based approaches for monitoring mobility, safety and activity in healthcare settings (Patel et al., 2012; Faragher and Harle, 2015). Most products focus on single functions and often generate frequent false alarms due to sensitivity to minor movements or environmental

interference (Liu et al., 2016). There are many systems for rehabilitation activity monitoring, physiological trends and movement detection (Patel et al., 2012), but the author has not found a system that integrates these in a dashboard.

Recent reviews show that motion-tracking technologies using accelerometers, gyroscopes, Artificial Intelligence (AI) and Internet of Things (IoT) are technically capable of capturing rich data on daily activity and behaviour. However, there remains a significant gap between the collection of motion data in controlled research environments and its effective translation into practical, real-world healthcare applications (Wei and Wang, 2024). Wearable activity trackers have also been explored in rehabilitation settings. Research shows that step counts, activity duration, heart rate and sleep data can provide meaningful indicators of recovery and engagement with therapy (Patel et al., 2012; Wright et al., 2017). Long-term monitoring of activity and behavioural patterns has been shown to help identify changes in functional status and health risk in older adults, supporting the clinical value of longitudinal data collection (Liu et al., 2016).

Overall, there is currently little evidence of integrated systems that combine wearable physiological data, indoor BLE movement detection and retrieval-augmented AI analysis within a single rehabilitation dashboard. This project seeks to explore that gap through a proof-of-concept implementation focused on clinically meaningful patient mobility monitoring.

FitBit Charge 6

There are a vast range of activity trackers available on the market, with some, such as the Apple Watch or the Oura rings, claiming to provide clinically validated information. For convenience, the author selected a model that he owns, a Fitbit Charge 6. This provides a range of information on the device itself and associated app. The data is accessible from the 'Fitbit Cloud' via an Application Programming Interface (API). It includes:

Clinical & Health Data

- ECG Waveform & Results
- Heart Rate Time Series
- Heart Rate Variability
- SpO2 (Daily Average & Nightly Levels)
- Breathing Rate (Sleep)

- Skin Temperature Variation
- Cardio Fitness Score
- Resting Heart Rate

Non-Clinical & Activity Data:

- Step Count
- Calories Burned
- Distance
- Floors/Elevation
- Active Zone Minutes
- Sleep Stages (Light, Deep, REM, Wake)
- Sleep Efficiency & Duration
- Exercise Logs (GPS coordinates, duration, and pace)

The data may not all be available on a continuous 'live' basis but can be pulled at user-determined intervals from the FitBit cloud, provided a device that has been set up with a unique user email. Some data may only be retrospective e.g. sleep data is only pushed to the cloud after a sleep episode and some requires permissions; this project focused on sample data.

Movement sensors

As part of the early exploration phase, a review was undertaken with the assistance of ChatGBT and Gemini to identify an appropriate movement detection sensor capable of reliably capturing initiation of movement within an indoor rehabilitation environment. The initial focus was on low-cost, body-worn devices that could detect changes in position and motion without requiring cameras. A key aim was to avoid false alarms that might arise from cameras or general sensors, as these might be triggered by staff or indeed other patients who might walk into a shared room. In addition, cameras of any type give rise to significant privacy concerns in a shared bedroom environment.

A range of options were considered, including simple accelerometer-based trackers, pressure-trigger devices, and Bluetooth Low Energy (BLE) movement sensors. While basic accelerometers are effective at detecting linear movement and step activity, they are more limited in identifying subtle postural transitions such as turning, leaning, or rising from a

seated or lying position. For this reason, a sensor incorporating a three-axis gyroscope in addition to tri-axial accelerometers was selected (Chen et al, 2023). The inclusion of gyroscopic data allows detection of rotational motion and orientation changes, which are particularly relevant in rehabilitation contexts where early movement initiation, such as sitting up, turning in bed, or standing. This combined sensing approach provides richer motion data than accelerometers alone and should offer a more robust basis for identifying clinically meaningful movement events. The selected device also supports BLE connectivity, allowing integration with a Raspberry Pi gateway for real-time data capture and transmission to the backend system, making it suitable for use in the planned proof-of-concept architecture.

The movement sensor selected was a 'WIT Motion WT901BLECL' as this combines tri-axial accelerometers with a three-axis gyroscope, enabling detection of both linear movement and rotational changes in body position while supporting real-time data transmission to a local gateway (Raspberry Pi) via Bluetooth. For this project, the sensor device was positioned at waist level, as arm movements could distort the readings. The optimum positioning of the sensors is probably a study in itself.

Results

The overall System Design and Architecture achieved is illustrated in Figure 1

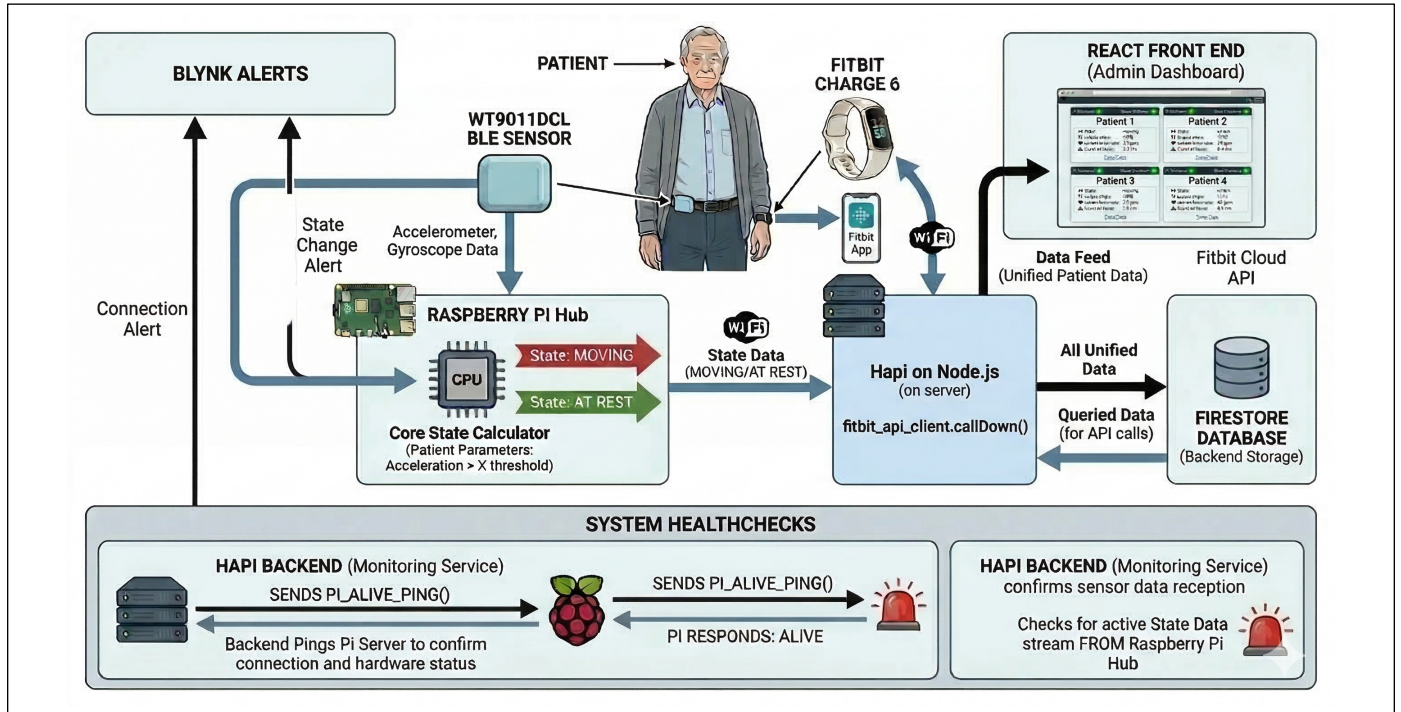


Figure 1 System Architecture

The final architecture adopts a distributed, per-patient model comprising the following components:

- A wearable device (Fitbit Charge 6), linked to a unique user account and mobile application, providing activity and physiological data via an API. A separate JSON-based process was developed to retrieve and persist historical data for each patient in Firestore. This enabled both real-time monitoring and retrospective time-series analysis, providing meaningful insights into patient activity levels and trends.
- A WIT Motion BLE inertial sensor, capturing tri-axial gyroscopic movement data.
- A Raspberry Pi 4 acting as an edge device, running three Python programs (two on distributed deployment):

- `run_sensor.py`: a configurable movement reporting service that processes real-time sensor data using the WIT BLE communication interface and classifies movement states and sends same to the backend.
- `heartbeat.py` (sensor heartbeat): monitors the times of flow of sensor data to determine whether the sensor is actively reporting to the backend
- Each pi programme has to be customised with the patient ID and the sensor MAC address. A corresponding ID has to be entered in the Firestore database.
- `pi_ping.py` (local deployment only): periodically transmits a heartbeat signal to the backend to confirm device availability, enabling differentiation between sensor failure (e.g. battery depletion) and device failure (e.g. power loss).
- A backend service implemented in Node.js using the Hapi framework, responsible for data ingestion, processing, and system monitoring.
- A `heartbeat.js` service within the backend that receives periodic status messages from each Raspberry Pi and records them both in memory and in Google Firestore. A detailed operational record is maintained in the devices collection, while key status indicators (including connection status and sensor data recency) are mirrored to the corresponding patient record. This supports persistent monitoring of device health and enables efficient retrieval of status data for presentation in the user interface.
- A cloud-based persistence layer using Google Firestore, integrating Cloudinary image resources
- A React-based frontend for visualisation, incorporating Recharts for dashboard presentation and Cloudinary for image delivery. Patient cards present real-time status indicators, including sensor state and connectivity, while individual dashboards provide time-series visualisations of activity and physiological data. Independent filtering controls (e.g. day, week, month) support longitudinal analysis, allowing clinicians to interpret both current status and historical trends.
- A push connection to the Blynk service for alerts on mobile device. This was given a specific trigger of 15 seconds of sustained movement after the patient is classed as being at rest; those details are configurable.

Results: Screen shots

The following screen shots illustrate key features of the system created:

Python programmes running on 2 Raspberry Pi's connected to back end

The figure displays four terminal windows arranged in a 2x2 grid, showing data from two Raspberry Pi sensors connected to a back-end system.

- Top-left window:** A terminal window titled "conor — conor@raspberrypi: ~/ProjectSETU/FinalProject/rehab-dashbo...". It displays a continuous stream of sensor data for two patients (p1 and p2). Each line shows acceleration (Gx, Gy, Gz) and magnetic field (Gmag) values, followed by a status (REST or MOVING). For example, "Gx: 0.00 Gy: 0.00 Gz: 0.00 | Gmag: 0.00 REST".
- Top-right window:** A terminal window titled "conor — old-pi@old-pi: ~/sensor2 — ssh old-pi@192.168.1.58 — 80x24". It displays similar sensor data for two patients (p2 and p1), with status updates like "STATE CHANGE → MOVING".
- Bottom-left window:** A terminal window titled "conor — conor@raspberrypi: ~/ProjectSETU/FinalProject/rehab-dashbo...". It shows the execution of a Python script "python heartbeat.py" and the resulting heartbeat signals. The output includes JSON-like structures with fields like "type", "piId", "patientId", "connected", "lastFrameAgeSec", and "timestamp".
- Bottom-right window:** A terminal window titled "conor — old-pi@old-pi: ~/sensor2 — ssh old-pi@192.168.1.58 — 80x24". It also shows the execution of "python heartbeat.py" and the resulting heartbeat signals, mirroring the structure in the bottom-left window.

Figure 2 Raspberry Pi Data flows including BLE sensors and heartbeat signals

Sample telemetry data received by backend

```
Telemetry processed for p2 via pi-p2: sample REST
Telemetry received: {
  type: 'sample',
  piId: 'pi-p1',
  patientId: 'p1',
  state: 'REST',
  gmag: 0,
  timestamp: '2026-03-29T13:22:53.024342+00:00'
}
evaluateMovementAlert {
  patientId: 'p1',
  prevState: 'REST',
  incomingState: 'REST',
  prevLastChangedAtMs: 1774790551484,
  prevRestQualified: false,
  prevMovementStartedAtMs: null,
  prevAlertSent: false,
  eventMs: 1774790573024
}
REST duration check {
  patientId: 'p1',
  restStartMs: 1774790551484,
  restDurationMs: 21540,
  REST_MIN_MS: 30000,
  prevRestQualified: false
}
```

Figure 3 Backend Console Data Flow

React Front End Welcome Page

Rehab Dashboard PoC
Welcome

Demo Mode

**Welcome to the
Rehab Dashboard Proof of
Concept**

Please confirm location and proceed to the dashboard.

Hospital
Select

Ward
Select

Username
Clinician 1

Password
.....

Click to Enter

Figure 4 React Welcome page

‘Ward’ Patient Care view prior to Pi activation. Note red warning lights and card colours to indicate ‘no connection’

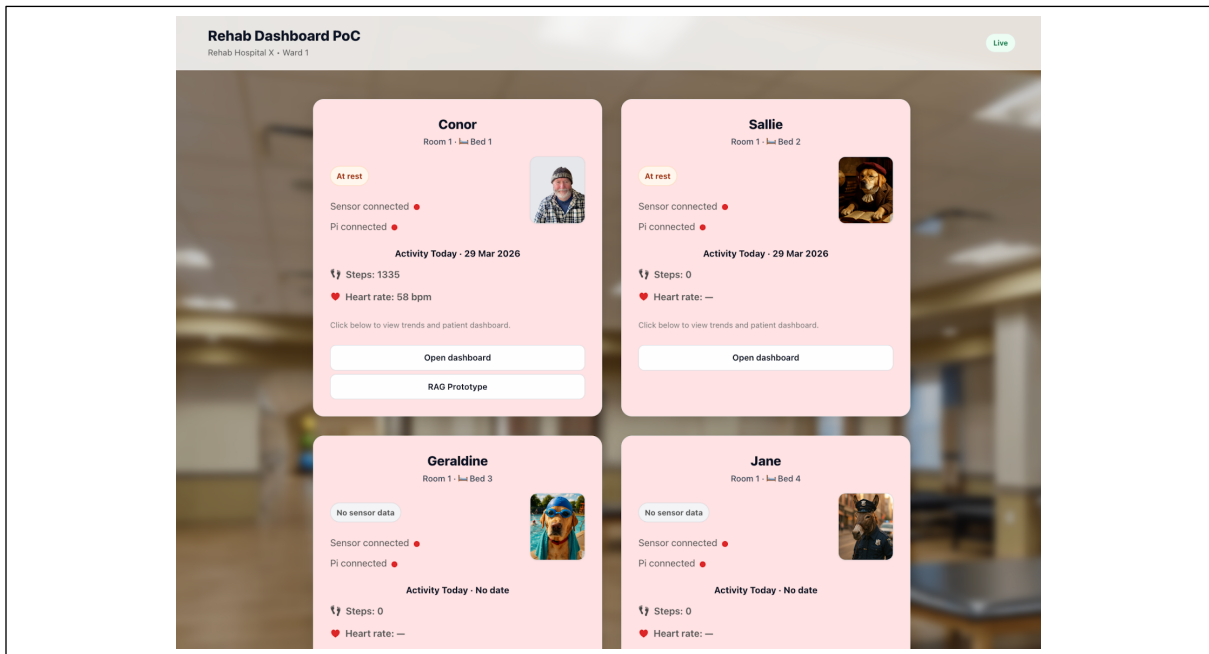


Figure 5 Pt Card prior to Pi activation

Ward view post 2xPi activated. Note change of colours in card and connection radio buttons

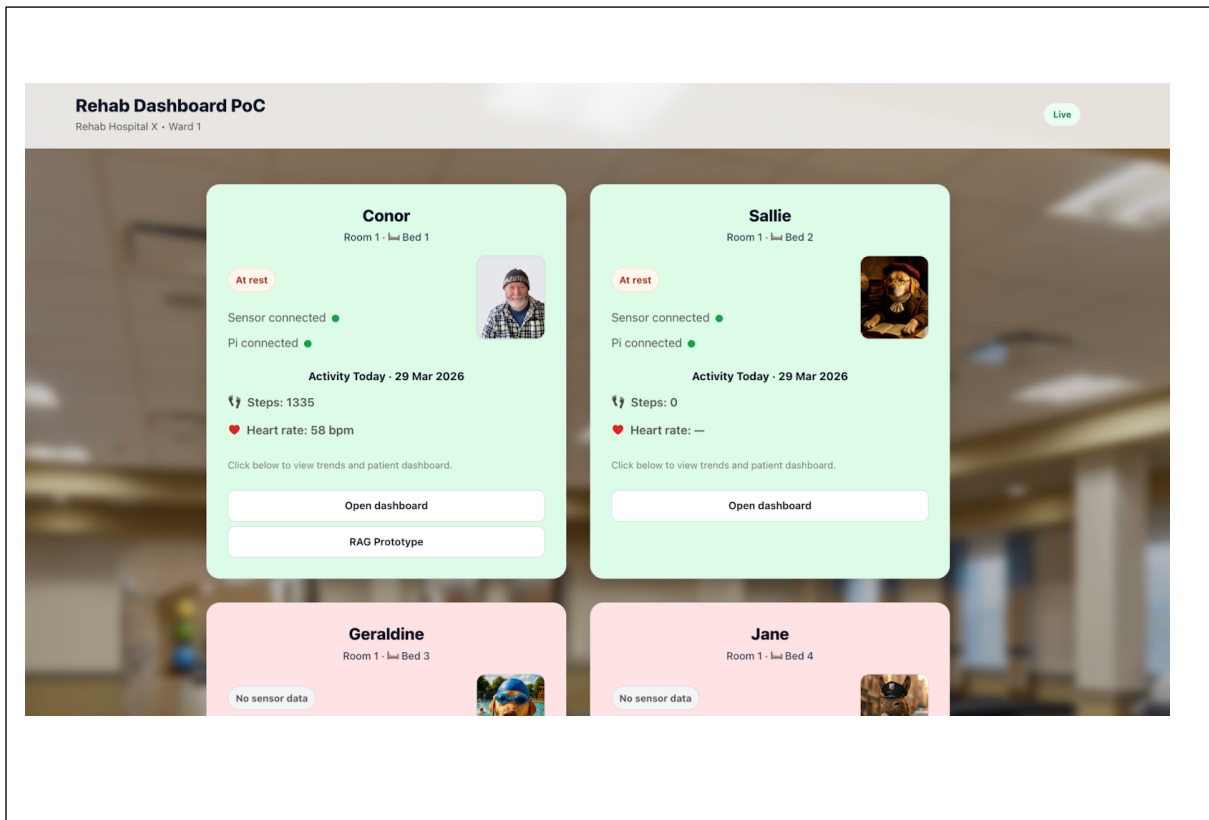


Figure 6 Patient card post activation

Alert for sustained movement after period of rest

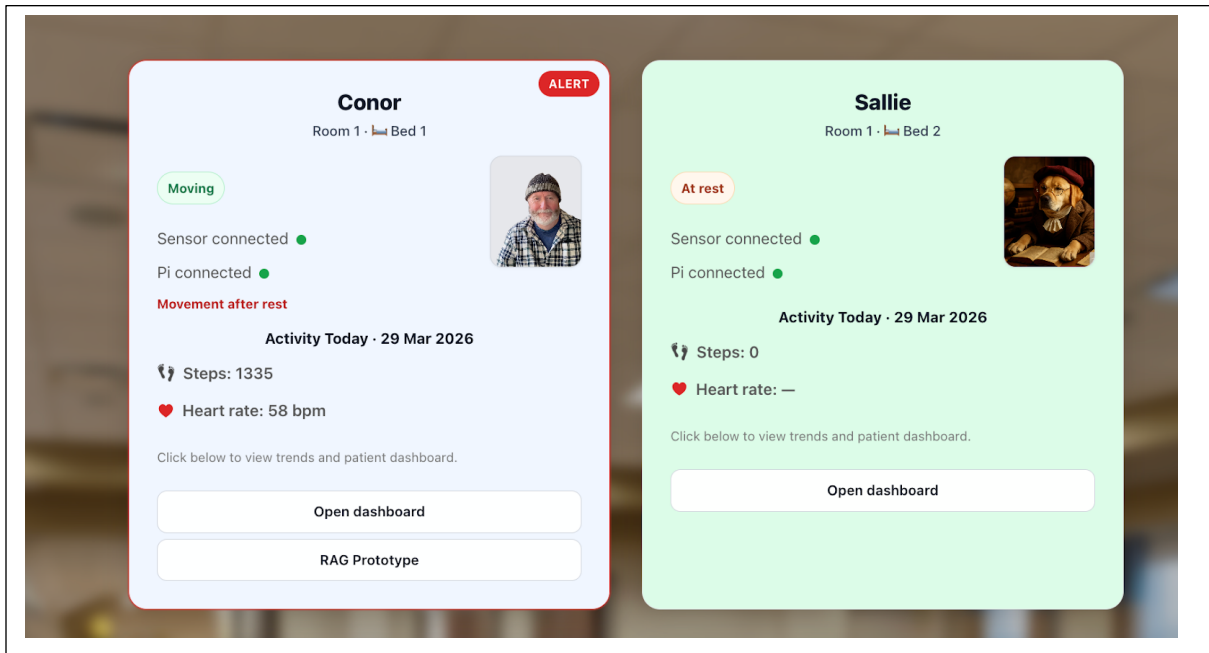


Figure 8 Alert Feature triggered by movement after rest

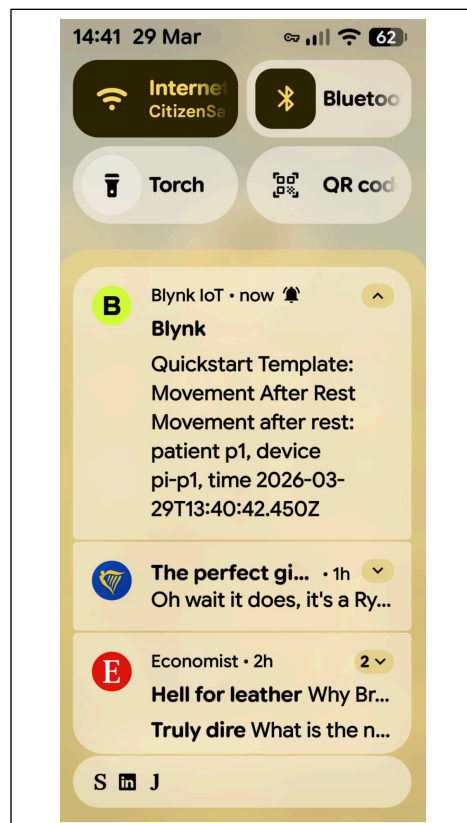


Figure 7 Mobile Device Blynk Alert

Section of Patient Dashboard View with Trend Filters

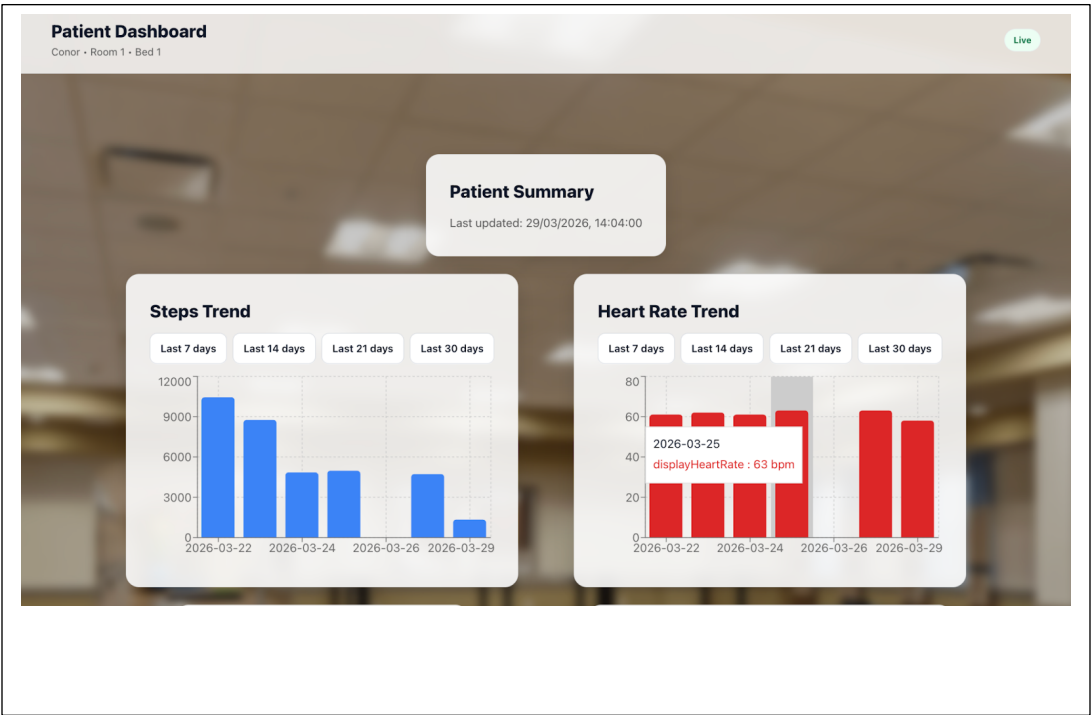


Figure 9 Sample dashboard view for patient

Implementation Learning/Reflection

IoT

As a Raspberry Pi Model 4 can support multiple blue-tooth connections, the author assumed it would be able to track a number of BLE sensors simultaneously, meaning that one Pi could potentially serve a multi-occupancy room on a ward. However, early testing quickly determined that the Pi could not sustain multiple connections whilst simultaneously monitoring the continuous stream of data emanating from a sensor.

In theory, a Pi could have ‘polled’ sensors in turn i.e. connect to sensor 1, review current data, classify, transmit and then move on to sensor 2 etc, but this would have resulted in periods when an individual sensor was not being monitored. The author concluded that this was not ideal in a clinical setting and instead a single sensor – Pi relationship would provide more robust, continuous monitoring.

This required reconfiguration of the code to allow for multiple Pi’s: each one must now have an identification code which can be transmitted to the backend, which in turn associates the data with the correct patient while monitoring multiple Pi data streams.

Movement detection and data persistence

The WT901BLECL is a multi-sensor inertial measurement unit (IMU) incorporating a tri-axial accelerometer, gyroscope and magnetometer. Through sensor fusion algorithms, it provides real-time measurements of acceleration, angular velocity, orientation (Euler angles) and magnetic field across three orthogonal axes, enabling comprehensive analysis of movement and posture in three-dimensional space.

RAG and movement classification

Following early examination of sample sitting to standing movements, the author concluded that a combined magnitude reading of the tri-axial gyroscope readings could give an initial quantification of rotational movement intensity. This could be used to broadly measure movement to support this PoC.

The movement magnitude was calculated as:

$$\omega_{mag} = \sqrt{\omega_x^2 + \omega_y^2 + \omega_z^2}$$

Equation 1

where $\omega_{x,y,z}$ represent angular velocity across three orthogonal axes. This approach provides an orientation-independent measure of movement (meaning the sensor can be attached in any position) and enabled differentiation between different movement patterns. While accelerometer-based magnitude is commonly used to detect overall movement intensity, gyroscope-derived magnitude provides additional insight into the rotational characteristics of movement (Chen et al).

There may well be more sophisticated approaches to the application and analysis of the movement data available, but this measure was selected as a starting point.

After the key aspects of the project has been addressed, the author explored the possibility of applying Retrieval Augmented Generation of movement classification. Json datasets were created for moving in bed, lying to sitting, sitting to standing and falls. The data was passed to ChatGPT for analysis. This identified that distinct patterns of movement were discernible, particularly for falls. While this is extremely interesting, a key aim of the project is to prevent the falls in the first place.

It was suggested that supplementing gyro readings with specific accelerometer readings could enhance the classification of movement, but this also would have meant mounting the sensors with a specific orientation. That is not likely to be sustainable in clinical settings.

It may be possible to use these readings to enhance the 'state' classification for individual patients i.e. instead of binary 'at rest' or 'moving', it may be possible to apply parameters for a third state e.g. 'standing' or 'sitting up'. These could be built into the python programme for an individual patient, thereby resulting in an entirely customisable monitoring service. Some preliminary information on this aspect was coded as a static page in the front end, purely for information purposes. This aspect is worthy of further study.

Sample movement graphs generated by ChatGBT

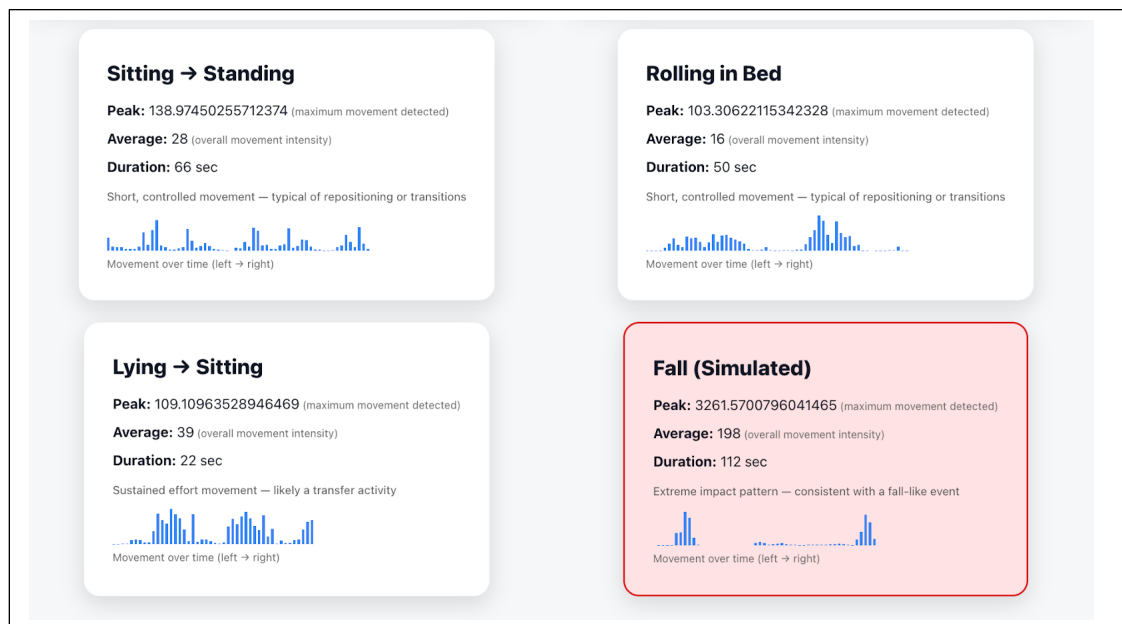


Figure 10 Exploring Movement Classification

Data Persistence

Given the continuous stream of data available from the sensor, a key decision required was storage of same. To avoid building up a vast database of sensor readings, it was instead decided to use the gyro-based magnitude readings as an initial means of classifying a state of being either 'at rest' or 'moving' with timestamps. Based on an analysis of the preliminary

sample data a threshold of 'movement magnitude' ≥ 8 was applied to the angular velocity magnitude calculation, to differentiate between the two states.

In the early stage of the project, the Pi was set to report every change of 'state', with a timestamp to allow analysis. To reduce storage, the programme was set to cease reporting state if there was a continuous period of rest of 5 minutes or more, until a new state of 'moving' was triggered. However, that resulted in no data going to the backend, with the outcome that one couldn't be sure if there was nothing to report or if no data was reaching the back-end. The python programme was modified to send state every 10 seconds or on any change.

Data presentation

Further work would be required to ensure that the front-end presentation means social inclusion standards e.g. use of screen readers by people with visual impairment.

Blynk

The IoT service 'Blynk' was configured to receive alerts. The limitations of a standard account meant that alerts to this service were not entirely reliable, due to built-in messaging frequency constraints coupled with fixed message quotas. This triggered a re-design of the patient card to make these responsive to changes in patient state.

Conclusion

The project demonstrates that integrating multi-source data streams into a clinician-facing dashboard is feasible, if somewhat constrained by a requirement for mobile phones, subscriber accounts and access permissions to data. An enterprise approach would probably require a separate, dedicated architecture, still built around Bluetooth for superior data flows. The identification of BLE constraints of the Raspberry Pi resulted in a more realistic architecture in this project, and despite the need for one pi per sensor, it could be extremely cost effective when one considers the implications of a serious patient fall.

In addition to cost, ease of use, reliability and accuracy of observations is essential in clinical systems. Significant work would be required to test this in a clinical environment. Future work could include full implementation of sensor analytics building on RAG, different alerting mechanisms, alternative architectures such as enterprise gateways for a BLE mesh. Clinical pilot testing is required to validate usability and impact.

The project delivered a viable PoC rehabilitation dashboard integrating wearable and sensor data, with a robust monitoring architecture and a scalable deployment model. It provides a strong foundation for further development, with possible future clinical testing.

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